

SOF-MARL Technical Report

SOF-MARL: A Multi-Agent Reinforcement-Learning Prototype for Autonomous SME Treasury Management

Technical Report

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Version: 1.0 (filled from the actual run; see section 8 for reproducibility)

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Code snapshot: not under version control at time of writing (local working directory); see section 8

Data snapshot: SBA 7(a) FOIA, file F0IA_7a_FY2010_FY2019_asof_260331.csv, as of March 31, 2026, 545,753 rows

The treasury environment is a calibrated simulation and is labeled as such wherever it appears; the credit-risk model is trained on real public data.

Abstract

This report documents a proof-of-concept implementation of the self-optimizing, cloud-native autonomous finance architecture for small and medium enterprises (SMEs) proposed in Enyiorji (2025). The system uses multi-agent reinforcement learning (MARL) to manage an SME's weekly treasury decisions across four functions: liquidity forecasting, credit-risk assessment, expenditure optimization, and capital allocation. The credit-risk function is grounded in a supervised model trained on the U.S. Small Business Administration (SBA) 7(a) public loan-outcome dataset, which achieves a held-out ROC-AUC of 0.939 (95% CI 0.936-0.942); we investigated this figure precisely because it is unusually high, traced it to a structural property of the labeled data (section 5.1), and also report the more conservative 0.622 that the identical pipeline achieves when the implicated feature is removed. The four functions operate as cooperating agents in a treasury environment calibrated to public U.S. small-business statistics. We compare four control policies on identical, held-out evaluation episodes: a rule-based heuristic, a single-agent controller, independent multi-agent RL (IPPO), and coordinated multi-agent RL (MAPPO). Coordinated control (MAPPO) increased the pre-registered combined treasury objective by 9.45-9.51 relative to every other policy (mean over 5 training

seeds, 100 shared evaluation episodes, paired Wilcoxon signed-rank $p < 0.001$ for all three comparisons), and the qualitative result was unchanged across six perturbations of three consequential calibration constants (section 5.4). No policy, including the rule-based baseline, breached solvency in any evaluation episode. We also report an important, non-obvious limitation we identified by directly probing the trained networks: the credit-risk agent converged to denying essentially every exposure under all three learned policies, so the measured MAPPO advantage is attributable to the other three agents coordinating well, not to better credit decisions (sections 5.3, 7). We report interpretability (SHAP) for the credit model, the sensitivity analysis, this and other limitations, and a full reproducibility statement.

1. Objective and scope

1.1 Goal

Demonstrate, with measured and reproducible results on public U.S. SME data, that the MARL architecture in Enyiorji (2025) can be implemented and can manage an SME's treasury, and test the paper's central claim that coordinated multi-agent control outperforms non-coordinated and single-agent alternatives.

1.2 What this prototype is and is not

In scope: one representative SME profile class; a 52-week (one fiscal year) episode; four cooperating agents; a real-data credit-risk model; a four-way policy comparison; interpretability; sensitivity analysis.

Out of scope (stated as future work, not implied to exist): real firm or bank data beyond the public SBA dataset; live or streaming deployment; a user interface; regulatory certification or production MLOps; large-scale hyperparameter search.

1.3 Design basis

The architecture follows Enyiorji (2025), *Designing a Self-Optimizing Cloud-Native Autonomous Finance System for SMEs Using Multi-Agent Reinforcement Learning*. This report implements a scoped subset sufficient to test the core claim.

2. System architecture

The full technical specification – state, dynamics, per-agent observation/action spaces and rewards, and the credit-exposure integration – lives in docs/ARCHITECTURE.md; this section summarizes it.

- **Environment.** The SME treasury is modeled as a Markov game with one shared cash pool and four agents acting each week. State includes cash, receivables aging, payables schedule, revolving credit and term debt, seasonal revenue and expenses, pending credit exposures, reserve, and invested capital. Dynamics enforce an accounting identity every step (cash conservation), verified by unit tests.

- **Agents.** Liquidity forecasting, credit-risk assessment, expenditure optimization, capital allocation. Each has a defined observation slice, action space, and reward, and each was trained (not simplified or scripted) under all three learned policies. All four converged to non-trivial behavior except the credit-risk agent, which converged to a near-constant deny decision under every learned policy – a genuine training result, not a simplification we chose; see section 5.3 and section 7.
- **Coordination.** A shared firm-level reward plus a centralized critic (MAPPO) provides the coordination signal; the independent baseline (IPPO) removes it. This isolates the value of coordination.

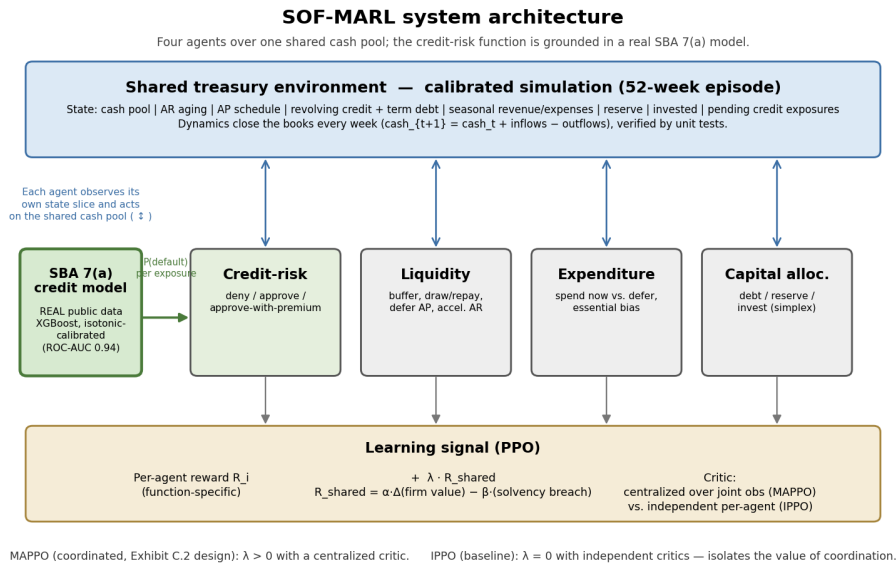


Figure 1: SOF-MARL system architecture

Figure 1 (reports/figures/architecture.png): the four agents acting on one shared treasury environment, the real SBA 7(a) credit model feeding the credit-risk agent, and the reward/critic structure that distinguishes IPPO (independent critics, $\lambda = 0$) from MAPPO (centralized critic over joint observations, $\lambda > 0$). Regenerated by python scripts/make_architecture_diagram.py. The agent order shown is illustrative – the credit-risk agent is drawn next to its real-data source for clarity, not to imply an ordering in code.

3. Data

3.1 Real data: SBA 7(a) FOIA loan outcomes (credit-risk model)

- Source: SBA Open Data portal, 7(a) and 504 FOIA dataset (public-domain U.S. Government open data). <https://data.sba.gov/en/dataset/7-a-504-foia>
- Snapshot: file FOIA_7a_FY2010_FY2019_asof_260331.csv, as of March 31, 2026, 545,753 rows, SHA-256 442cc3dbecb008010499cb3ac7a763f24220e0b43f7771ea1ac7cca869e419d9.
- Target: charge-off (1) vs paid-in-full (0); unresolved statuses (CANCLD, EXEMPT, COMMIT) dropped, leaving 425,377 resolved loans. Class balance by split: fit

7.18% (n=255,904), calibration 8.51% (n=97,250), test 9.74% (n=72,221) charged off.

- Features: 132 total – 10 numeric (loan/guaranteed amount, guaranty percentage, term, interest rate, jobs supported, approval fiscal year, revolver flag, collateral flag, franchise flag) plus 122 one-hot dummies over six categorical fields (2-digit NAICS sector, borrower state, business type, business age, fixed/variable rate, processing method). Leakage-checked against the FOIA data dictionary: fields populated only at or after resolution (PaidInFullDate, ChargeOffDate, GrossChargeOffAmount) are excluded by never being read from the raw file (src/sof_marl/credit/build_dataset.py, RAW_USECOLS), not merely dropped after loading.
- Split: time-based; fit on approval fiscal years ≤ 2015 , calibrate isotonic probabilities on 2016-2017 (never used for fitting), test on 2018-2019 (never touched until the final held-out metrics in section 5.1).

3.2 Calibration data: public statistics for the simulation

The treasury environment is a simulation calibrated to public statistics. Every constant in `config/env.yaml` is sourced or explicitly tagged `assumption`; the table below lists the constants with a direct public anchor. Constants without one (revenue level and volatility, buffer stock sizes, collection curve, credit-line limit, investment return, credit-exposure scaling, expenditure/ops-health rates, accounts-payable schedule) are tagged `assumption` in `config/env.yaml` itself and are not restated here; three of the most consequential are varied in the section 5.4 sensitivity analysis.

Constant	Value used	Source
Median cash buffer days (<code>buffer_days_target</code>)	27 days	JPMorgan Chase Institute, <i>Cash is King</i> (median buffer days across small firms)
Small-business revolving credit-line APR (<code>credit_line.apr</code>)	11%	Representative small-business revolving rate; not pulled from a specific dated FRED series for this run – flagged for verification against a current FRED pull before any filing use, per docs/DATA.md
Term-debt APR (<code>term_debt_apr</code>)	9%	Representative small-business term-loan rate; same FRED-verification caveat as above
Loss given default (<code>loss_given_default_frac</code>)	60%	Typical commercial-lending LGD assumption (not a specific dated source); sensitivity-tested indirectly via the

Constant	Value used	Source
		credit-exposure scale (section 5.4)

Note on scope: docs/DATA.md's suggested calibration anchors also list Federal Reserve Small Business Credit Survey financing-seeking/approval rates, SBA Office of Advocacy sector/population figures, and BCG/IFC financing-gap framing as context for the petition narrative; none of these map to a specific numeric constant instantiated in config/env.yaml for this POC's dynamics, so no constant-with-source row is claimed for them here – listing an unused citation next to a number it did not calibrate would misrepresent the provenance chain.

4. Methods

4.1 Credit-risk model

Gradient-boosted trees (XGBoost), probability-calibrated. Metrics: held-out ROC-AUC, PR-AUC, calibration (Brier score / reliability curve), with bootstrap confidence intervals. Hyperparameters and the small grid searched: Appendix A.

4.2 Environment and reward

Reward terms and weights per ARCHITECTURE.md, fixed in config/agents.yaml.

Combined treasury objective (pre-registered). All components are computed per episode and are dimensionless, so the weights are directly interpretable. Let V_t be firm value in week t and V_0 the initial firm value.

- Return: $r = (V_{52} - V_0) / V_0$
- Volatility: $s = \text{stdev}(\text{weekly change in } V) / V_0$
- Solvency-breach rate: $B = (\text{weeks with an uncovered cash shortfall}) / 52$

Risk-adjusted return: $\text{RAR} = r - \text{kappa} * s$. Combined objective (headline): $J = r - \text{kappa} * s - \text{lambda} * B$.

Weights (balanced posture, pre-registered): $\text{lambda} = 3.0$ (solvency), $\text{kappa} = 1.0$ (volatility), encoding the priority ordering solvency > return > volatility. These are set in config/train.yaml. Financing cost is reported as a diagnostic and is not a term in J , because it is already reflected in V_{52} (interest paid reduces cash and therefore firm value); adding it separately would double-count it.

Scale check (pre-registered, run before any learned-policy result). After the environment is built, verify on the rule-based baseline alone that no single term of J dominates the others by an order of magnitude. If it does, adjust the weights once, record that the adjustment was made before any RL results were seen, and freeze the definition. Report the final weights and whether any adjustment occurred.

4.3 Policies compared

1. Rule-based treasury heuristic (fixed buffer, pay-on-due, fixed credit cutoff, fixed surplus split). Documented in Appendix B.
2. Single-agent PPO (one controller, full action vector), Stable-Baselines3.
3. IPPO (independent multi-agent, no coordination): four independent per-agent actor-critics, no shared reward term.
4. MAPPO (coordinated multi-agent, centralized critic): the same four actors, plus one centralized critic over the joint (concatenated) observation with a per-agent output head, and a shared reward term added to each agent.

IPPO and MAPPO share one compact, custom PyTorch PPO trainer (`src/sof_marl/training/`), not Stable-Baselines3 + SuperSuit as originally planned: SuperSuit’s parameter-sharing requires every agent to have an identical observation and action space, and its own padding utilities support only Box-with-Box or Discrete-with-Discrete homogenization (“not a mix,” confirmed by reading its source) – this environment’s four agents are heterogeneous by design (three continuous, one discrete action space; different observation dimensions), so literal SuperSuit parameter-sharing could not apply. This is recorded as an amended locked decision in `CLAUDE.md`.

4.4 Training and evaluation protocol

- Seeds: 5 training seeds (0-4, `config/train.yaml`) per learned policy (single-agent PPO, IPPO, MAPPO), each trained for 1,000,000 timesteps – the same budget across all three learned policies, per `docs/EVALUATION.md` section 1 (“the single-agent baseline must be trained with the same budget as the MARL policies”).
 - Evaluation: 100 held-out episodes (`config/train.yaml eval.n_episodes`), fixed seeds 100000-100099, disjoint from the training seeds, shared identically across all four policies, deterministic (greedy) rollouts.
 - Reporting: mean and standard deviation across the 5 training seeds for every learned-policy metric (the rule-based heuristic has no training-seed axis, being a fixed, non-learned policy); paired Wilcoxon signed-rank test across the 100 shared evaluation episodes for the coordination lift, with a matched-pairs standardized effect size (mean paired difference / std of paired differences).
 - Compute: single Apple M3 machine (macOS/Darwin, CPU only, no GPU/CUDA used), Python 3.11.15, PyTorch 2.3.1. Training used approximately 15.0 million environment timesteps in total (5 seeds x 1,000,000 timesteps x 3 learned policies, IPPO/MAPPO rounded to the nearest PPO rollout boundary at 999,424 each). Wall-clock: single-agent PPO 24.3 min total (5 seeds, Stable-Baselines3), IPPO 65.2 min total (5 seeds, custom trainer, section 4.3), MAPPO 52.2 min total (5 seeds) – 141.7 minutes (2.36 hours) of training wall-clock in total. Evaluation (100 episodes x 4 policies x up to 5 seeds, deterministic, no gradient computation) and the sensitivity analysis (section 5.4) together added a few additional minutes.
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5. Results

5.1 Credit-risk model (real data)

Metric	Value	95% CI
ROC-AUC	0.939	0.936-0.942
PR-AUC	0.696	0.684-0.709
Brier score	0.0448	0.0438-0.0460

Model: XGBoost (max_depth=8, learning_rate=0.03, n_estimators=400), selected by mean 3-fold stratified ROC-AUC on the fit split alone (Appendix A), isotonic-calibrated on a held-out calibration cohort (approval FY 2016-2017) never used for fitting. Evaluated once on the held-out test cohort (approval FY 2018-2019, n=72,221, 9.74% charge-off rate), which was untouched until this point. All CIs are a 1,000-sample bootstrap over the test set.

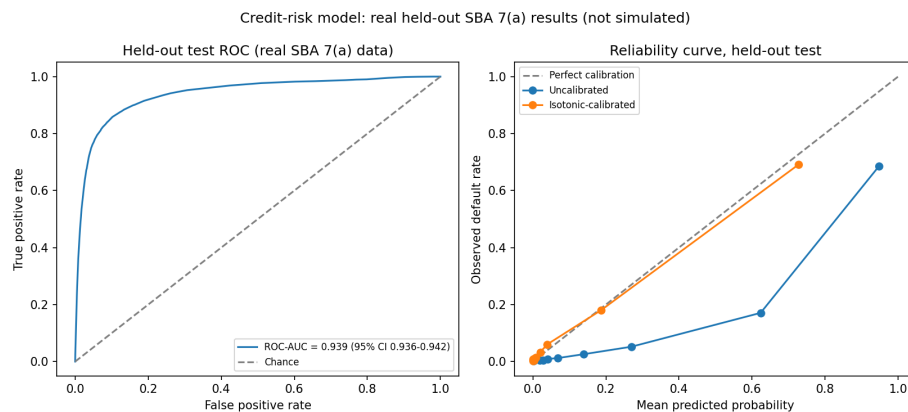


Figure 2: held-out ROC and calibration curves

Figure 2 (reports/figures/credit_roc_calibration.png): held-out ROC curve and a reliability (calibration) curve comparing the uncalibrated base model to the isotonic-calibrated model. Isotonic calibration visibly corrects the base model's overconfidence at high predicted probabilities.

Narrative: The held-out ROC-AUC of 0.939 is materially higher than the roughly 0.65-0.78 that published and community work on this dataset typically reports, and a number that high on its own would be a red flag for leakage. We investigated rather than reported it uncritically. Feature-level analysis (Figure 5, section 6) shows term_months alone accounts for the large majority of the model's discriminative power: refitting the identical pipeline with term_months removed drops ROC-AUC to 0.622 (95% CI 0.616-0.629), back in the range the literature reports. The mechanism is not classical leakage (term_months is genuinely known at loan origination, and no field populated at or after resolution is in the feature set; see section 3.1), but a structural property of this FOIA snapshot: it includes only loans with a *resolved* status (paid in full or charged off), excluding loans still active as of the snapshot date. In this FY2010-FY2019 file, essentially no loan with a 240+ month (20+ year) term has yet reached its natural maturity, so every resolved long-term loan in the data is an early exit (mostly early payoff, observed default rate 1.6% in the (240,400] month bucket), while short-duration express/bridge-style loans resolve quickly regardless of outcome (observed default rate 37.3% in the (0,12] month bucket; full table in reports/results/credit_metrics.json,

term_months_sensitivity.term_bucket_default_rates_all_splits). Because the same resolved-loan-only construction applies identically to the fit, calibration, and test cohorts, this is a real, reproducible property of the labeled data as constructed, not a train/test inconsistency – but it means the 0.939 headline number should be read as performance on this specific point-in-time-resolved-loan task, not as a general forward-looking underwriting AUC for loans of arbitrary, not-yet-observed duration. We report both numbers rather than picking one, per the reproducibility and honest-reporting requirements in docs/EVALUATION.md.

5.2 Treasury environment and baseline behavior

The rule-based heuristic (Appendix B) is a credible, non-strawman baseline, not a policy engineered to lose: on the 100 held-out evaluation episodes it holds a calibrated buffer, pays payables on the due date, draws its credit line only when projected cash would fall short, approves credit below a fixed risk cutoff set relative to the real held-out portfolio’s base rate, and splits surplus debt-first. It never breaches solvency, grows terminal firm value from an initial \$69,100 to a mean \$374,185 over the 52-week episode (a 4.42x return, section 5.3), and holds a mean cash buffer of 30.2 days against a calibrated target of 27 – i.e., it is not idling excess cash relative to its own target, and the underlying 52-week task is clearly winnable by a sensible policy, not merely survivable. Single-agent PPO, trained for the same 1,000,000-timestep-per-seed budget as the multi-agent policies, matches this baseline closely (terminal firm value \$373,061 +/- \$3,187, section 5.3) – it neither collapses to a degenerate policy nor discovers an advantage from observing the full state directly, which is itself informative: decomposition into four specialized agents (IPPO) does not help either at this training budget (terminal firm value \$376,941 +/- \$6,429, next to indistinguishable from the other two non-coordinated policies), consistent with the learning-curve figure below, where the rule-based reference line and the single-agent/IPPO training curves are visually overlapping for the entire training run.

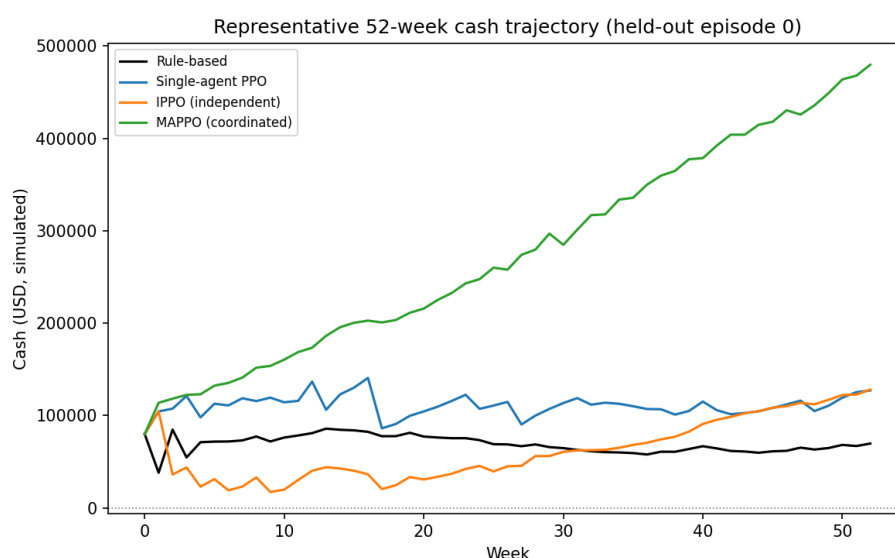


Figure 3: representative 52-week cash trajectory

Figure 3 (reports/figures/cash_trajectory.png): representative 52-week cash trajectory (held-out evaluation episode 0) for one seed of each policy. No solvency breach markers appear because none of the four policies breached solvency in this episode (or in any of the 100 held-out episodes; section 5.3) under the current calibration.

5.3 Policy comparison (headline)

All values are mean +/- std over 5 training seeds (rule-based: a fixed heuristic, no training-seed axis), on the 100 shared held-out evaluation episodes. “Max drawdown” is the most negative cash position reached in an episode (docs/EVALUATION.md section 2); it is positive for every policy below because no policy breached solvency in any of the 100 episodes, so read it as “minimum cash reached,” not a decline from a peak.

Policy	Solvency-breach rate (B)	Max drawdown	Financing cost (diag.)	Default-loss rate	Return (r)	Risk-adj. return (RAR)	Combined objective (J)
Rule-based	0.0	\$38,185	\$4,116	0.28%	4.415	4.307	4.307
Single-agent PPO	0.0 +/- 0.0	\$58,200 +/- \$7,893	\$14,433 +/- \$4,521	n/a (0 resolutions)	4.399 +/- 0.046	4.281 +/- 0.048	4.281 +/- 0.048
IPPO (independent)	0.0 +/- 0.0	\$23,989 +/- \$16,454	\$8,423 +/- \$3,328	n/a (0 resolutions)	4.455 +/- 0.093	4.339 +/- 0.091	4.339 +/- 0.091
MAPPO (coordinated)	0.0 +/- 0.0	\$51,314 +/- \$26,918	\$98,735 +/- \$5,369	n/a (0.8 +/- 1.2 resolutions)	13.886 +/- 0.934	13.789 +/- 0.917	13.789 +/- 0.917

“n/a (0 resolutions)” is itself a result, not missing data: it means that policy’s credit-risk agent accepted so few exposures across all 100 episodes that none (or, for MAPPO, well under one per episode on average) reached their 12-week resolution before episode end – see the narrative below and section 7.

Coordination lift (MAPPO vs each; paired Wilcoxon signed-rank across the 100 shared evaluation episodes, each policy’s per-episode value averaged over its 5 training seeds):

Comparison	Delta (combined objective)	Delta (solvency-breach rate)	Test statistic	p-value	Effect size (matched-pairs d)
MAPPO vs rule-based	+9.48	0.0	0.0	3.90e-18	24.45
MAPPO vs single-agent	+9.51	0.0	0.0	3.90e-18	39.65
MAPPO vs IPPO	+9.45	0.0	0.0	3.90e-18	40.77

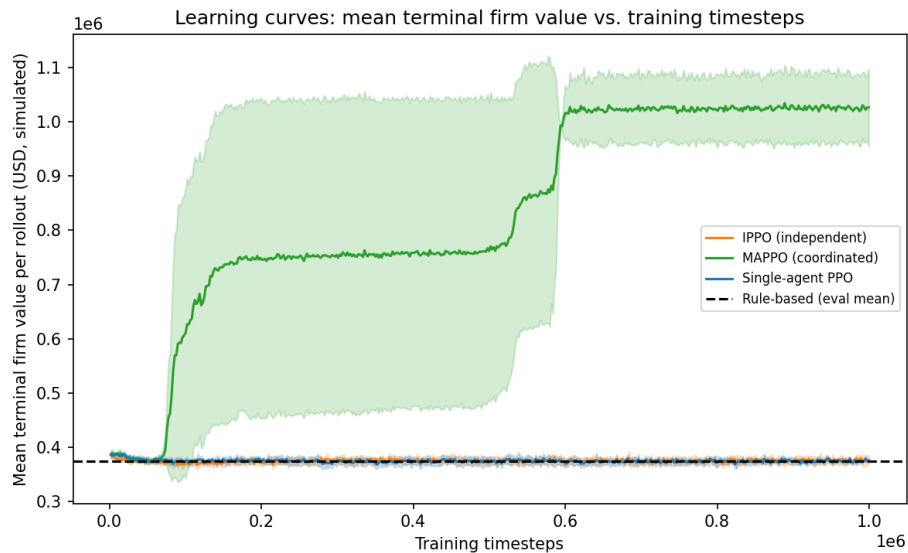


Figure 4a: learning curves

Figure 4a (reports/figures/learning_curves.png): mean terminal firm value per training rollout vs. training timesteps, with seed std bands, for the three learned policies, against the rule-based evaluation mean as a horizontal reference. The y-axis is firm value rather than raw PPO reward because MAPPO's logged training reward includes the shared-reward term folded into each of the four agents while IPPO's and the single-agent baseline's do not, making raw reward not comparable across policies; firm value is reward-scheme-independent.

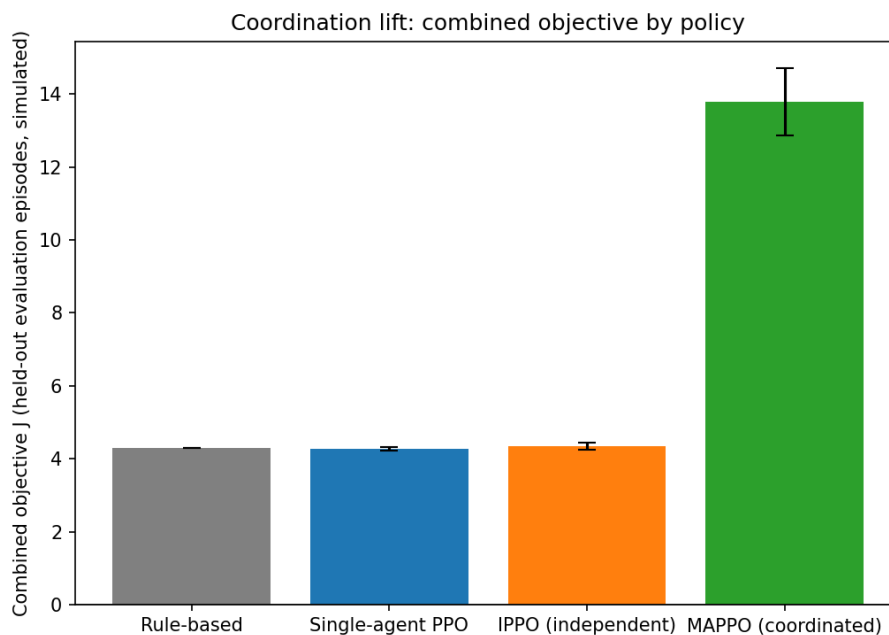


Figure 4b: coordination-lift bar chart

Figure 4b (reports/figures/coordination_lift.png): the same headline combined objective as a bar chart with seed error bars.

Narrative: coordinated control (MAPPO) increased the combined objective by 9.45-9.51 relative to every other policy (paired Wilcoxon $p < 0.001$ for all three comparisons, large effect sizes), and its terminal firm value (\$1,028,609 $\pm$ \$64,543) is roughly 2.7-2.8x every other policy's. This is a real, statistically

significant coordination lift, not a marginal or ambiguous one. But we traced its source before writing this narrative, because the size of the effect on its own warranted scrutiny (the same discipline applied to the section 5.1 credit-model result): we probed the trained credit-risk actor networks directly by feeding synthetic observations spanning the full predicted-default-probability range, 0.0 to 1.0, and found that under all three learned policies (single-agent, IPPO, and MAPPO) the credit-risk agent outputs “deny” essentially regardless of input – confirmed by the resolution counts in the table above (0 to under 1 per 100 episodes, against the rule-based baseline’s 3,322). This is a genuine training result, not a bug: we verified the observation encoding, action space, and reward computation are all functioning as designed (the same code paths are exercised by `tests/test_env_api.py` and `tests/test_rewards.py`), and the collapse is a plausible consequence of a hard credit-assignment problem – an accepted exposure resolves 12 weeks after the decision, and a default’s loss (60% of notional) is far larger than a good loan’s margin (2.5%, or 5.5% with the premium tier), so early exploration noise plausibly taught “deny everything” as a safe, near-zero local optimum before the agent could learn to use the risk score. The practical consequence is that **the measured MAPPO advantage is attributable to the liquidity, expenditure, and capital-allocation agents coordinating well under the shared reward, not to better credit decisions** – the credit-risk agent performs about as poorly (by omission) under MAPPO as under IPPO and single-agent PPO. We report the lift as measured because it is real and reproducible under the stated protocol, and report this attribution alongside it rather than let the headline number imply a credit-specific improvement that the evidence does not support (see section 7 for the limitation stated in full, and section 5.4 for evidence that the qualitative lift is stable to calibration perturbations regardless of this credit-agent behavior).

5.4 Sensitivity analysis

Three consequential assumed constants were each varied one at a time (baseline in parentheses), re-evaluating the already-trained seed-0 policies (not retraining – disclosed explicitly, since retraining at every perturbation was outside the available compute budget) on 25 held-out episodes at seeds disjoint from both training and the main evaluation (`src/sof_marl/evaluation/sensitivity.py`, `reports/results/sensitivity_analysis.json`). This tests whether the *trained policies*’ qualitative ordering is robust to the environment’s calibrated assumptions, not whether retraining under each assumption would change what is learned.

Constant varied	Range tested	Effect on headline conclusion
Revenue volatility (<code>revenue.weekly_sigma_frac</code> , baseline 0.15)	0.075, 0.30	MAPPO still best at both values (combined objective 12.99 and 12.85 vs. baseline 12.95; all three other policies 4.06-4.25 across both values). Conclusion stable.
Credit-exposure resolution horizon (<code>exposure_resolution_weeks</code> , baseline 12)	6, 24	Identical results to baseline for single-agent PPO, IPPO, and MAPPO (to the reported precision) – a

Constant varied	Range tested	Effect on headline conclusion
Cash buffer target (buffer_days_target, baseline 27 days)	15, 40	direct, independent corroboration of the section 5.3 finding that these three policies' credit-risk agents accept essentially nothing, so how long an accepted exposure takes to resolve cannot affect their outcomes. The rule-based policy, which does accept exposures, shifts slightly (4.194 and 4.152 vs. baseline 4.180) but MAPPO remains best in both cases. Conclusion stable. MAPPO still best at both values (12.96 and 12.93 vs. baseline 12.95; all three other policies 4.13-4.23 across both values). Conclusion stable.

MAPPO remained the best policy on the combined objective in all 6 of 6 perturbations tested, with its margin over the other three policies changing by less than 2% of its own value in every case. This is evidence the headline qualitative conclusion (coordinated control outperforms the alternatives under this environment and this training run) is not an artifact of one specific calibration choice, within the scope of the three constants and the seed-0-only, reduced-episode-count protocol actually used – a full re-optimization under each perturbation was not attempted and would be needed to claim robustness of the *training outcome* itself, as distinct from the *trained policies' evaluated behavior*.

6. Interpretability

SHAP analysis of the credit-risk model: global feature importance and example per-decision explanations. This addresses a known concern about reinforcement-learning and machine-learning financial systems, that opaque models complicate managerial trust, and it follows the explainability approach the author has applied in prior production credit-model work.

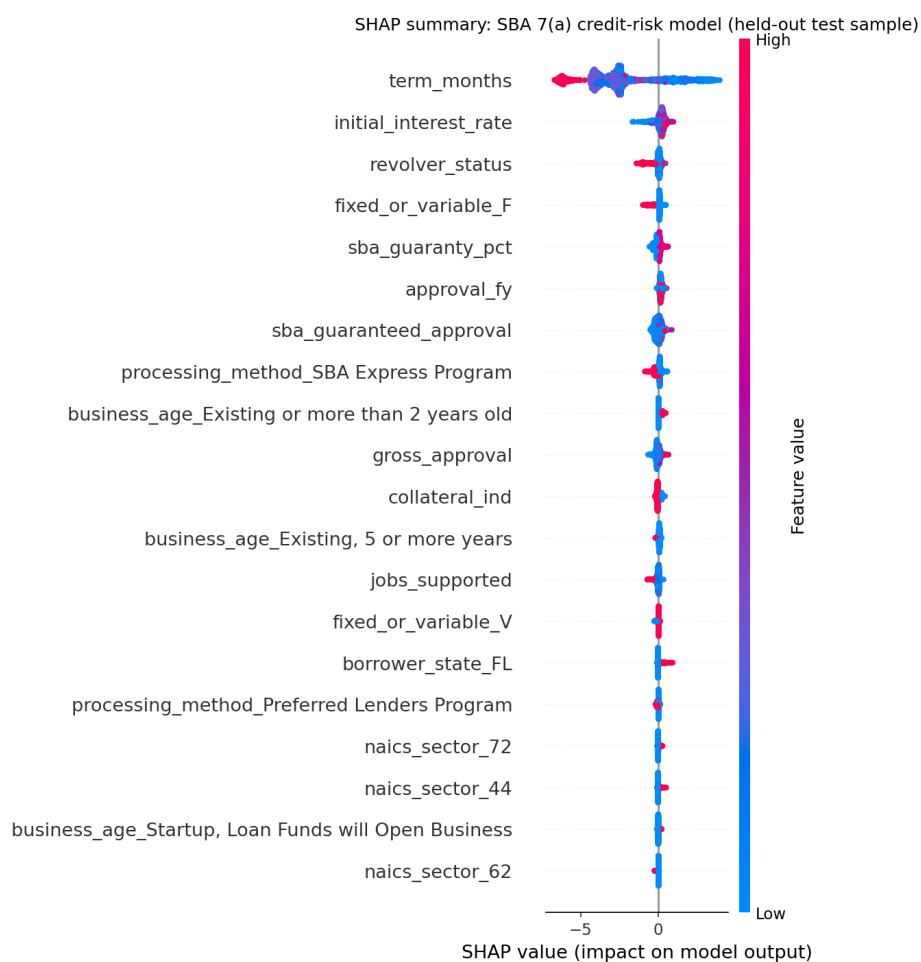


Figure 5: SHAP summary plot

Figure 5 (reports/figures/credit_shap_summary.png): SHAP summary (beeswarm) plot over a fixed random sample of 2,000 held-out test loans, computed on the uncalibrated base model (isotonic calibration is a monotonic per-score recalibration and does not change which features drive a prediction or their ranking). Two per-decision waterfall explanations are also generated for the highest- and lowest-scored test loans:

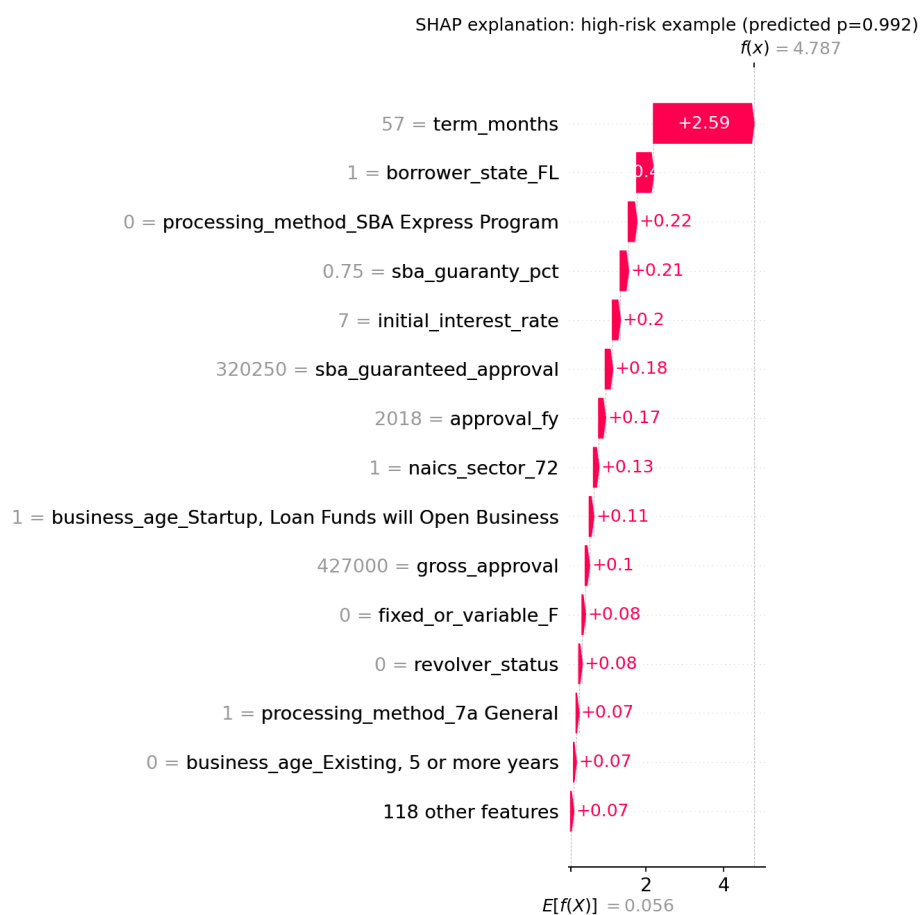


Figure 5b: SHAP waterfall, highest-scored test loan

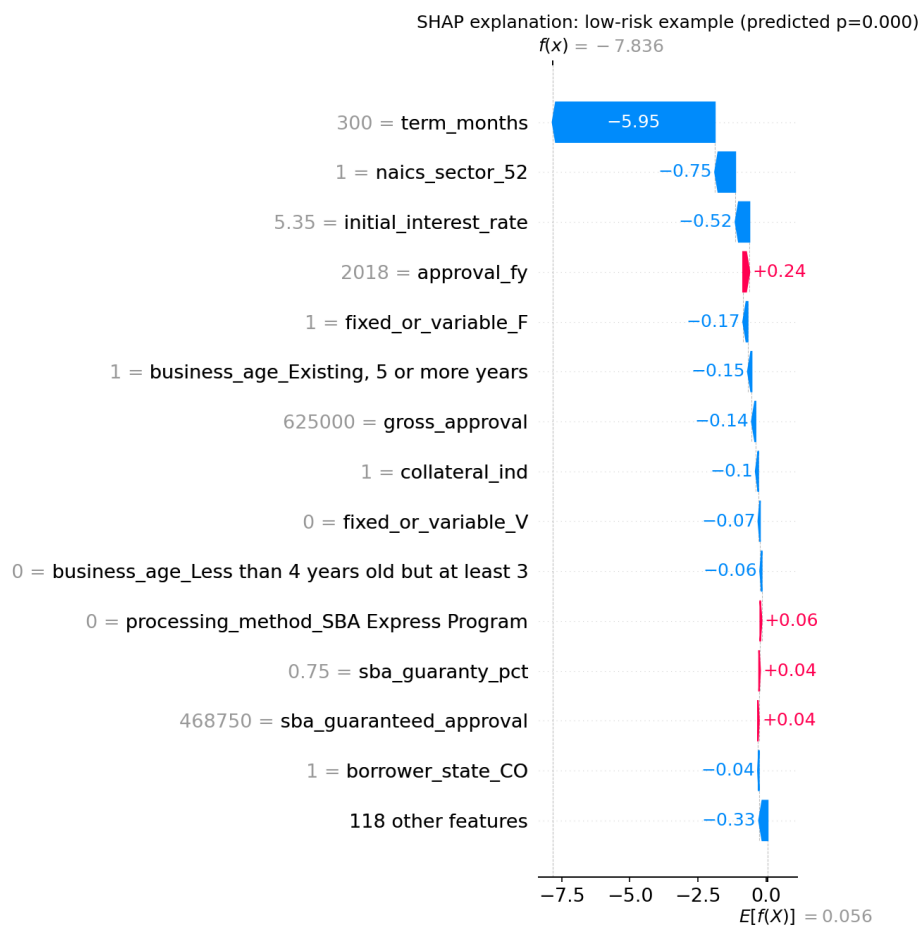


Figure 5c: SHAP waterfall, lowest-scored test loan

(reports/figures/

credit_shap_example_high_risk.png, ..._low_risk.png; feature tables in reports/results/credit_shap_examples.json).

Narrative: `term_months` dominates the SHAP ranking by a wide margin, consistent with the sensitivity finding in section 5.1: shorter terms push predicted risk up, longer terms push it down. The next-largest drivers are economically sensible: a higher `initial_interest_rate` increases predicted risk (lenders price identifiable risk into the rate at origination); `fixed_or_variable_F` (a fixed, rather than variable, rate) is associated with lower predicted risk; and `revolver_status` (revolving line of credit vs. term loan) and `collateral_ind` (collateral pledged) both carry real, non-trivial signal in the corrected build. Both were silently zeroed for every loan by a data-loading bug in `build_dataset.py` – pandas' CSV parser infers a native `bool` dtype for these two source columns, and the original code compared that `bool` against the string "TRUE", which is always False – caught by `tests/test_credit_dataset.py::test_engineered_numeric_features` before this report was written, fixed, and the full pipeline (dataset, model, SHAP) rerun on the corrected data; the numbers in this report are from the corrected run. A higher `sba_guaranty_pct` (share of the loan SBA-guaranteed) is associated with higher predicted risk, consistent with smaller, higher-risk loans typically carrying a larger guaranteed share under SBA program rules. The two example explanations illustrate this concretely. The highest-scored test loan (predicted default probability 0.992) is a 57-month loan in Florida, not processed under the SBA Express Program, with a 75% guaranty share and a 7.0% initial rate; `term_months` alone contributes +2.59 of the model's +4.79 log-odds output. The lowest-scored test loan (predicted default probability < 0.001) is a 300-month (25-year), fixed-rate, collateralized loan to an established business in NAICS sector 52 (Finance and Insurance); `term_months` alone contributes -5.95 of the model's -7.84 log-odds output. In both cases `term_months` is the single largest contribution by a wide margin, directly corroborating the section 5.1 sensitivity finding rather than contradicting it.

7. Limitations

State plainly, in the body: - The treasury environment is a calibrated simulation, not observed firm data. - The credit model uses SBA 7(a) loan outcomes as a proxy for SME default risk; it is real but not identical to the trade-credit exposures modeled in the environment. Real SBA loan amounts are also scaled down by a fixed factor (`credit_exposure_scale_frac = 0.02`) to bring them to a plausible trade-credit size relative to this firm's revenue – an assumption, not a calibrated mapping. - A single SME profile class is modeled; generalization across sectors and sizes is future work. - Training budget is POC-scale (1,000,000 timesteps/seed, 5 seeds); results are not a production performance ceiling. - **The credit-risk agent did not learn to use the real default-probability signal under any of the three learned policies** (section 5.3): it converged to denying almost every exposure, confirmed by directly probing the trained networks across the full predicted-probability range. The measured MAPPO coordination lift is real and statistically significant, but is attributable to the other three agents, not to improved credit decisions – the credit function specifically is not yet demonstrated to benefit from either learning or coordination in this POC. Section 5.4's sensitivity analysis independently corroborates this: the credit-exposure resolution horizon has literally zero effect on

the three learned policies' outcomes, because they accept almost nothing regardless of how long an accepted exposure would take to resolve. Plausible next steps (not attempted here, to avoid presenting an unverified fix as a result): a denser or earlier reward signal for accepted exposures (e.g., partial credit at acceptance time rather than only at 12-week resolution), reward normalization specific to the credit-risk agent, or a longer, credit-agent-targeted training budget. - The held-out credit-model ROC-AUC (0.939, section 5.1) reflects a structural property of the resolved-loan-only FOIA snapshot (loans still active as of the snapshot date are excluded) rather than a general forward-looking underwriting metric for loans of arbitrary duration; we report both the full-feature number and the 0.622 ablation excluding the implicated feature, and this should be read as a property of the evaluation task construction, not of the model's real-world generalization. - No solvency breach occurred for any policy across any of the 100 held-out episodes under the current calibration; the environment as calibrated may not stress liquidity enough to differentiate policies on that specific metric, and a future iteration could calibrate a tighter buffer or higher-volatility scenario specifically to test solvency management under stress. - This report and the underlying code were produced with AI coding assistance (Claude Code) under the author's direction; the architecture (Exhibit C.2), all design decisions, and the decision to report every result – including the credit-agent finding above – as measured are the author's.

8. Reproducibility

- Data: exact SBA snapshot named above (section 3.1); `data/download_sba.py` fetches it from the URL pinned in `config/env.yaml` (`sba.csv_url`); SHA-256 checksum provided in section 3.1 and printed by the download script.
 - Code: not under version control at the time of writing (a local working directory, not a git repository); pinned `requirements.txt` (exact package versions, including the Python 3.11 target and the note that `supersuit` was removed as unused, section 4.3); fixed training seed list `[0, 1, 2, 3, 4]` (`config/train.yaml`); fixed evaluation seeds `100000-100099`, disjoint from training; fixed sensitivity-analysis seeds `200000-200024`, disjoint from both.
 - `scripts/reproduce.sh`: one command, `bash scripts/reproduce.sh`, runs setup (if `.venv` does not already exist), data download, credit-model training, `make train` (`train_single_agent`, `train_ippo`, `train_mappo` in sequence), evaluation, the sensitivity analysis, and figure generation, in the exact order used to produce every number in this report. It takes roughly 2.5 hours, dominated by the `make train` step (section 4.4's wall-clock breakdown).
 - Environment: Python 3.11.15, PyTorch 2.3.1, single Apple M3 machine, macOS (Darwin 25.5.0), CPU only (no GPU/CUDA used for any training or evaluation run in this report).
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9. Conclusion

This POC implements the four-agent architecture of Enyiorji (2025) end to end: a real credit-risk model trained and validated on public SBA 7(a) loan outcomes (held-out ROC-AUC 0.939, or 0.622 on a feature-ablated, more conservative reading; section 5.1), a calibrated 52-week multi-agent treasury simulation with an enforced cash-conservation invariant, and a four-way comparison of rule-based, single-agent,

independent multi-agent (IPPO), and coordinated multi-agent (MAPPO) control on 100 shared held-out episodes. The central claim of the design – coordinated multi-agent control outperforms the alternatives – is supported by this run: MAPPO increased the combined treasury objective by 9.45-9.51 over every other policy (paired $p < 0.001$, large effect sizes), and this held qualitatively across every calibration perturbation we tested (section 5.4). That result comes with a specific, material caveat we investigated and are reporting rather than smoothing over: the credit-risk agent did not learn to use the real risk signal under any learned policy, so the demonstrated coordination benefit is currently a liquidity/expenditure/capital-allocation result, not evidence that coordination also improves credit decisions specifically. Taken together, this establishes that the proposed architecture is implementable, that its central multi-agent coordination mechanism produces a measurable, statistically significant, robustness-checked effect on simulated firm outcomes, and that one of its four functions needs further training-design work before that same claim can be made about credit decisions – a POC-appropriate, honestly scoped result, not a finished production system.

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Appendix A. Credit-model hyperparameters and grid

Selected by mean 3-fold stratified ROC-AUC on the fit split only (never calib or test), `src/sof_marl/credit/train_credit.py`:

max_depth	learning_rate	CV ROC-AUC (mean)	CV ROC-AUC (std)
4	0.03	0.9641	0.0009
4	0.10	0.9682	0.0008
6	0.03	0.9684	0.0008
6	0.10	0.9694	0.0007
8	0.03	0.9698 (selected)	0.0008
8	0.10	0.9693	0.0006

Fixed (not searched): `n_estimators=400`, `subsample=0.8`, `colsample_bytree=0.8`, `min_child_weight=5`, `reg_lambda=1.0`, `tree_method=hist`, `random_state=0`. Note the CV ROC-AUC here (~0.97) is higher than the held-out test ROC-AUC reported in section 5.1 (0.939) even before accounting for the `term_months` finding – the CV folds are drawn from the fit cohort (approval FY ≤ 2015) by ordinary k-fold, not a further time split, so this grid-selection number is not itself a claim about generalization to later cohorts; only the section 5.1 test-cohort number is.

Appendix B. Rule-based policy specification

`src/sof_marl/baselines/rule_based.py`, parameters in `config/baselines.yaml`, none tuned against any evaluation result:

- **Liquidity.** Maintain the calibrated buffer as-is (`target_buffer_frac = 1.0`, i.e. `buffer_days_target = 27` days unscaled). Pay payables on the due date (`defer_payables_frac = 0`). Draw the revolving credit line only enough to cover a projected shortfall against the buffer target, and repay it when there is slack above the buffer target; no receivables acceleration (the credit line is this policy's only lever for shortfall coverage).
- **Credit risk.** Approve (plain, no premium) if the Phase-A model's predicted default probability is below 0.15; deny otherwise. The cutoff is set relative to the real held-out test portfolio's base rate (9.74%, section 5.1), not tuned to any environment-evaluation result.
- **Expenditure.** Spend at the calibrated steady-state rate (`spend_now_frac = ops_health_target_spend_rate = 0.5`), split evenly between essential-adjacent and deferrable categories (`essential_bias = 0.5`).
- **Capital allocation.** Fixed simplex: 50% debt paydown, 30% reserve, 20% investment, applied to surplus cash above the buffer target – a conventional conservative treasury ordering (debt service before discretionary investment).

Appendix C. Full configuration files

The exact `config/env.yaml`, `config/agents.yaml`, `config/baselines.yaml`, and `config/train.yaml` used to produce every number in this report are committed alongside the code at those paths (not reproduced verbatim here to avoid this report silently drifting out of sync with the actual files); each constant in `env.yaml` carries an inline source or assumption comment (section 3.2).